ROSENBERG INDEX, RICCI-FLATNESS AND PARALLEL SPINORS ON FINITE COVERINGS

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Editorial scope. Only TheoremA is counted. Official native SIGMA TeX and its complete four-step proof are retained unchanged, including author macros and full bibliography. Special-holonomy corollary is uncounted. Rendering omits journal frontmatter/class and provides an empty LastPageEnding journal-footer macro; no mathematical changes.

Allowed standard prerequisites (editorial).

- Bourguignon scalar-curvature rigidity and Ricci-flat splitting. Kazdan1975a; author Theorem3.1 with Fischer1975 and Cheeger1971 references.
- Finite-dimensional detection of higher-index spectrum. Schick2021a Proposition3.6, reproduced exactly as author Proposition3.2.
- Schroedinger-Lichnerowicz formula, spinor curvature and covering classification. Friedrich2000, Lawson1989, Hatcher2001; exact author uses and bibliography retained.

1.Introduction and main results

Let (M, g) be a connected Riemannian manifold of dimension n whose universal cover $\tilde{M}$ admits a spin structure. We denote by $\tilde{g}$ the pullback metric of g along the covering map $\tilde{M} \to M$. If $\left(\tilde{M}, \tilde{g}\right)$ carries a parallel spinor, g is Ricci-flat, i.e., its Ricci tensor vanishes identically. The converse of this statement holds for closed connected Riemannian spin manifolds with non-zero $\hat{A}$-genus or more general non-zero Hitchin invariant [10, Section 4.2]. This is a direct consequence of the Schrödinger–Lichnerowicz formula and the non-triviality of the kernel of the spin Dirac operator. It is an open question whether the converse of this statement holds in general.

Question 1.1. *If g is Ricci-flat, does $\left(\tilde{M}, \tilde{g}\right)$ admit a non-trivial parallel spinor?*

In this note, we give a positive answer to Question 1.1 for closed connected spin manifolds with non-vanishing Rosenberg index (see Theorem A). As a consequence, in dimension ≥ 2 these manifolds have special holonomy for any Ricci-flat metric (see Corollary 1.3).

The Rosenberg index of a closed spin manifold is a class in the real K-theory of the maximal group C*-algebra of the fundamental group of the manifold. It is defined as the higher index of the $\mathbb{C}\mathrm{l}_n$-linear spin Dirac operator twisted by the Mishchenko bundle (see Section 2). In general, the Rosenberg index is not only non-vanishing whenever the $\hat{A}$-genus or the Hitchin invariant is non-zero, but also for many other classes of manifolds, as for example enlargeable manifolds [7, 8], and aspherical manifolds whose fundamental group satisfies the Novikov conjecture [17].

Theorem A. *Let (M, g) be a closed connected Riemannian spin manifold with non-vanishing Rosenberg index. Then the following statements are equivalent:*

(1) *The Riemannian manifold (M, g) is Ricci-flat.*

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(2) *The universal cover $\tilde{M}$, equipped with the pullback metric, admits a non-trivial parallel spinor.*
(3) *There exists a finite Riemannian covering carrying a non-trivial parallel spinor with respect to the pullback spin structure.*
(4) *The metric g has non-negative scalar curvature.*

The main part of the proof of Theorem A is to show the implication (4)⇒(3). The proof strategy is as follows. Suppose there exists a non-trivial twisted harmonic spinor. Then, by the Schrödinger–Lichnerowicz formula and the assumption on the scalar curvature, this twisted spinor is already parallel. We obtain a non-trivial parallel spinor on the universal covering of M, which yields on a suitable finite covering a non-trivial parallel spinor with respect to the pullback spin structure. In the following paragraph, we outline the construction of such a twisted harmonic spinor under the assumption (4) in Theorem A.

A non-zero classical index such as the $\hat{A}$-genus or the Hitchin invariant gives rise to a non-trivial harmonic spinor. In general, a non-vanishing higher index such as the Rosenberg index does not give rise to a non-trivial harmonic spinor. But non-vanishing of the Rosenberg index is still an obstruction — so far the best obstruction based on the index theory of Dirac operators — to the existence of a positive scalar curvature metric on a closed connected spin manifold. A classical rigidity statement by Bourguignon states that every closed Riemannian manifold that does not admit a positive scalar curvature metric is Ricci-flat if the scalar curvature is non-negative (see, e.g., [11]). Combining these two facts yield under assumption (4) that (M, g) is Ricci-flat. By the *structure theorem for Ricci-flat manifolds* (see Theorem 3.1), the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index, and it is possible to detect the non-vanishing Rosenberg index by the spin Dirac operator twisted by some flat finite-dimensional Hermitian vector bundle (see Proposition 3.2). This detection principle we are using here is due to Schick and Wraith [18], while an alternative approach is given in [15, Section 3]. Here Ramras, Willett, and Yu [15] study the class of groups for which an associated non-vanishing higher index can be detected by a finite representation.

We now discuss the relation between Theorem A and special holonomy. A Riemannian manifold is called *irreducible* if the representation of the reduced holonomy group on the general linear group of the tangent space is irreducible. By Berger's holonomy list [1], the reduced holonomy group of any connected irreducible Ricci-flat manifold of dimension n is given by $\mathrm{SO}(n)$ in the generic case, or one of the following subgroups: $\mathrm{SU}(n/2)$ (locally Calabi–Yau), $\mathrm{Sp}(n/4)$ (locally hyper-Kähler), G_2 or $\mathrm{Spin}(7)$. In dimension ≥ 2, the assumption that the Riemannian manifold is irreducible excludes all flat manifolds, in particular all Ricci-flat locally symmetric spaces [2, Theorems 10.72 and 7.61]. Note that the subgroups $U(n)$ (locally Kähler) and $\mathrm{Sp}(n/4)\cdot\mathrm{Sp}(1)$ (locally quaternionic Kähler) cannot occur as holonomy groups of Ricci-flat manifolds [2, Proposition 10.29 and Theorem 14.45]. While there exist examples of closed locally Calabi–Yau, locally hyper-Kähler, G_2, and $\mathrm{Spin}(7)$ manifolds, no closed irreducible Ricci-flat manifold with generic reduced holonomy is known. This leads to the following question.

Question 1.2. *Does every closed connected Ricci-flat manifold of dimension $n \geq 2$ have special holonomy, i.e., its reduced holonomy group is a proper subgroup of $\mathrm{SO}(n)$?*

The *de Rham splitting theorem* states that every complete simply-connected Riemannian manifold is isometric to a Riemannian product of complete non-flat irreducible manifolds and a Euclidean space [2, Theorem 10.43]. We obtain that for reducible manifolds and flat manifolds the conclusion in Question 1.2 holds.

Since every Ricci-flat manifold of dimension ≤ 3 is already flat, this includes all manifolds of dimension 2 and 3. In dimension ≥ 4, Question 1.2 is still open. It is related to spin geometry as follows. Every complete simply-connected irreducible Riemannian spin manifold $\big(\tilde{M}, \tilde{g}\big)$ admits a parallel spinor if and only if it has special holonomy (see [10, Theorem 1.2] and [20]). We obtain by Theorem A the following corollary, which addresses Question 1.2 in the special case of spin manifolds with non-vanishing Rosenberg index.

Corollary 1.3. *Every closed connected Ricci-flat spin manifold of dimension ≥ 2 with non-vanishing Rosenberg index has special holonomy.*

2.The Rosenberg index

In this section, we give a brief introduction to higher index theory with the goal to define the Rosenberg index [17]. See [13] for the construction of the ($\mathbb{C}\mathrm{l}_n$-linear) spinor bundle, [12] and [4] for some background concerning Hilbert C*-modules and differential operators acting on Hilbert C*-module bundles, [19, Definition 2.1] for a definition of graded real Dirac bundles carrying a Hilbert C*-module structure, and [16] for an introduction to real K-theory of C*-algebras and in particular for a definition of the spectral picture of real K-theory.

Let (M, g) be a closed connected Riemannian spin manifold of dimension n. This means that the manifold is oriented and equipped with a spin structure $\mathrm{Spin}(M, g)$, which is a lift of the $\mathrm{SO}(n)$-principal bundle of oriented orthonormal frames along the double covering $\mathrm{Spin}(n) \to \mathrm{SO}(n)$. The spin structure gives rise to the following two complex vector bundles

$$\begin{aligned} &\mathcal{S} \coloneqq \mathrm{Spin}(M, g) \times_{\rho} \Delta, && \rho\colon\ \mathrm{Spin}(n) \to \mathrm{Aut}(\Delta) && \text{complex spin representation,} \\ &\mathfrak{S} \coloneqq \mathrm{Spin}(M, g) \times_{\mathrm{cl}} \mathbb{C}\mathrm{l}_n, && \mathrm{cl}\colon\ \mathrm{Spin}(n) \to \mathrm{Aut}(\mathbb{C}\mathrm{l}_n) && \text{left multiplication.} \end{aligned}$$

They are called the *irreducible spinor bundle* and the $\mathbb{C}\mathrm{l}_n$*-linear spinor bundle*, respectively. Here we denote by $\mathbb{C}\mathrm{l}_n$ the complexification of the Clifford algebra of $\mathbb{R}^n$ which carries the structure of a graded real C*-algebra and a graded real Hilbert $\mathbb{C}\mathrm{l}_n$-module. The irreducible spinor bundle $\mathcal{S}$ equipped with its natural metric, connection and Clifford multiplication has the structure of a Dirac bundle. The $\mathbb{C}\mathrm{l}_n$-linear spinor bundle $\mathfrak{S}$, equipped with the grading, real structure and Hilbert module structure obtained fiber-wise from $\mathbb{C}\mathrm{l}_n$, the induced $\mathbb{C}\mathrm{l}_n$-linear connection, and the Clifford multiplication, carries the structure of a graded real $\mathbb{C}\mathrm{l}_n$-linear Dirac bundle. Note that $\mathfrak{S}$ is isomorphic as a Dirac bundle to a direct sum of copies of the irreducible spinor bundle $\mathcal{S}$.

We denote by π the fundamental group of the manifold M and by $\mathrm{C}^*\pi$ the maximal group C*-algebra of π. The *Mishchenko bundle* $\mathcal{L}$ is the associated bundle of the universal cover of M by the representation $l\colon \pi \to \mathrm{Aut}(\mathrm{C}^*\pi)$ given by left multiplication [14]. It carries a natural flat connection, a $\mathrm{C}^*\pi$-valued metric and a real structure. Tensoring the $\mathbb{C}\mathrm{l}_n$-linear spinor bundle by the Mishchenko bundle yields a graded real $\mathbb{C}\mathrm{l}_n \otimes \mathrm{C}^*\pi$-linear Dirac bundle. The induced $\mathbb{C}\mathrm{l}_n \otimes \mathrm{C}^*\pi$-linear Dirac operator $\mathfrak{D}_{\mathcal{L}}$ extends to a self-adjoint and regular unbounded operator

$$\mathfrak{D}_{\mathcal{L}}\colon\ \mathrm{H}^1(M, \mathfrak{S} \otimes \mathcal{L}) \longrightarrow \mathrm{L}^2(M, \mathfrak{S} \otimes \mathcal{L}).$$

Here the Sobolev spaces $\mathrm{H}^1(M, \mathfrak{S} \otimes \mathcal{L})$ and $\mathrm{L}^2(M, \mathfrak{S} \otimes \mathcal{L})$ are countably generated Hilbert $\mathbb{C}\mathrm{l}_n \otimes \mathrm{C}^*\pi$-modules which are similarly defined as the corresponding Sobolev spaces in the classical case. Since the manifold M is compact, the generalized Rellich theorem [4, Theorem 2.33] yields that the functional calculus for unbounded regular and self-adjoint operators on Hilbert C*-modules [4, Theorem 1.19] restricts to

$$\Phi_{\mathfrak{D}_{\mathcal{L}}}\colon\ \mathrm{C}_0(\mathbb{R}) \longrightarrow \mathrm{Kom}_{\mathbb{C}\mathrm{l}_n \otimes \mathrm{C}^*\pi}\big(\mathrm{L}^2(M, \mathfrak{S} \otimes \mathcal{L})\big). \tag{2.1}$$

This is stated in [4, Proposition 2.34] with $g \equiv 1$ and X a point. Here $\mathrm{Kom}_{\mathcal{A}}(E)$ denotes the compact operators on a Hilbert $\mathcal{A}$-module E for a C*-algebra $\mathcal{A}$. The map $\Phi_{\mathfrak{D}_{\mathcal{L}}}$ defines a class in the spectral picture of the n-th real K-theory group of $\mathrm{C}^*\pi$ (see, e.g., [16, Lecture 13]). It is called the *Rosenberg index* of the manifold M.

3.Proof of the main theorem

In this section, we give a proof of Theorem A. It is based on the structure theorem for Ricci-flat manifolds [5, Theorem 4.5], which is a direct consequence of the Cheeger–Gromoll splitting theorem [3], and a detection principle for the spectrum of Hilbert $\mathrm{C}^*\pi$-modules by Schick and Wraith [18].

Theorem 3.1 (structure theorem for Ricci-flat manifolds)**.** *Every closed connected Ricci-flat manifold M admits a finite normal Riemannian covering $N \times T^q \to M$ consisting of a closed simply-connected Ricci-flat manifold N and a flat torus T^q. In particular, the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index.*

Proposition 3.2 ([18, Proposition 3.6])**.** *Let π be a group containing the subgroup $\mathbb{Z}^q$ with finite index d, and let $\mathcal{B}$ be a possibly unbounded self-adjoint and regular operator on a countably generated Hilbert $\mathrm{C}^*\pi$-module. Suppose $T^2 - 1$ is compact in the sense of Hilbert $\mathrm{C}^*\pi$-modules for the bounded transform $T := \mathcal{B}\big(\mathcal{B}^2+1\big)^{-1/2}$. If 0 is in the spectrum of $\mathcal{B}$, then there exists a representation $\rho\colon \pi \to U(d)$ such that 0 is in the spectrum of $\mathcal{B} \otimes_\rho \mathrm{id}$.*

Proof of Theorem A. Let (M, g) be a closed connected Riemannian spin manifold of dimension n with non-vanishing Rosenberg index. We denote as in Section 2 the fundamental group of M by π, the universal cover of M by $\tilde{M}$, the Mishchenko bundle by $\mathcal{L}$, the irreducible spinor bundle by $\mathcal{S}$, the $\mathbb{C}\mathrm{l}_n$-linear spinor bundle by $\mathfrak{S}$, and the induced Dirac operators by $\mathcal{D}$ and $\mathfrak{D}$, respectively. Moreover, we equip the universal cover of M by the pullback metric $\tilde{g}$ and denote its irreducible spinor bundle by $\tilde{\mathcal{S}}$.

Since a parallel spinor on a finite covering of M lifts to a parallel spinor on the universal covering of M and the scalar curvature is by definition the trace of the Ricci curvature, the implications (3)⇒(2) and (1)⇒(4) are trivial. Suppose there exists a non-trivial parallel spinor u on $\tilde{M}$. The curvature tensor K of the spinor bundle $\tilde{\mathcal{S}}$ is related to the Ricci curvature via

$$\mathrm{Ric}_{\tilde{g}}(X) \cdot u = -2 \sum_{j=1}^{n} e_j \cdot K_{X,e_j} u \tag{3.1}$$

for any local orthonormal frame $e_1, \dots, e_n$ of $T\tilde{M}|_U$ and any vector field X supported in an open subset U [6, Section 3.1]. Since u is non-trivial and parallel, this yields that $\tilde{M}$ is Ricci-flat, hence M is Ricci-flat. This proves the implication (2)⇒(1). It remains to show the implication (4)⇒(3). Suppose the scalar curvature of the Riemannian manifold (M, g) is non-negative. We have to show that there exists a finite Riemannian covering of M together with a parallel spinor with respect to the pullback spin structure. The proof splits into the following four steps.

Step 1: The Riemannian manifold (M, g) is Ricci-flat. Since the Rosenberg index does not vanish, the manifold M does not admit a positive scalar curvature metric. Since the scalar curvature of M is non-negative, it follows by Ricci flow or a classical result by Bourguignon [11] that the manifold M is Ricci-flat. A new proof using only spinorial techniques is given by the author in [19, Sections 3 and 4] and will be briefly sketched here.

The non-vanishing Rosenberg index gives rise to a family of almost $\mathfrak{D}_{\mathcal{L}}$-harmonic sections $\{u_\epsilon\}_{\epsilon>0}$, i.e., $u_\epsilon \in \mathrm{C}^\infty(M, \mathfrak{S}\otimes\mathcal{L})$, $\|u_\epsilon\|_{\mathrm{L}^2} = 1$ and $\|\mathfrak{D}_{\mathcal{L}}^j u_\epsilon\|_{\mathrm{L}^2} < \epsilon$ for all $j > 0$ and all $\epsilon > 0$ [19, Lemma 3.2]. Since the scalar curvature of M is non-negative, we obtain by the Schrödinger–Lichnerowicz formula $\|\nabla u_\epsilon\|_{\mathrm{L}^2} < \epsilon$ for all $\epsilon > 0$. We can improve this estimate by Moser iteration [19, Lemma 3.4 ff.] and obtain $\|\nabla u_\epsilon\|_\infty \lesssim \epsilon^r$ for all $\epsilon \in (0,1)$ and a suitable positive constant r which is independent of ϵ. By the Poincaré inequality, we obtain that the family is almost constant, i.e.,

$$\big\|\bar{u}_\epsilon - \big\langle u_\epsilon(p), u_\epsilon(p)\big\rangle\big\|_{\mathbb{Cl}_n\otimes\mathrm{C}^*\pi} \lesssim \epsilon^r \qquad \text{with} \quad \bar{u}_\epsilon \coloneqq \frac{1}{\mathrm{vol}(M)}\int_M \big\langle u_\epsilon(q), u_\epsilon(q)\big\rangle \,\mathrm{d}q,$$

for all $\epsilon \in (0,1)$ and all $p \in M$. Furthermore, we can choose the family $\{u_\epsilon\}_{\epsilon>0}$ such that $\big\|\nabla^2 u_\epsilon\big\| \lesssim \sqrt{\epsilon}$ holds for all $\epsilon > 0$ [19, Lemma 4.3]. Together with equation (3.1), we obtain

$$\|\mathrm{Ric}(X)\|_{\mathrm{L}^2} \lesssim \|\mathrm{Ric}(X)\cdot u_\epsilon\|_{\mathrm{L}^2} \lesssim \sqrt{\epsilon} \tag{3.2}$$

for any vector field X on M and all sufficiently small $\epsilon > 0$, hence M is Ricci-flat.

Step 2: There exists a flat Hermitian bundle over M carrying a non-zero twisted parallel spinor. Since the manifold M is Ricci-flat by Step 1, we can apply Theorem 3.1 and the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index d for some q and d. The Rosenberg index of M does not vanish by assumption, hence the twisted Dirac operator

$$\mathfrak{D}_{\mathcal{L}}\colon\ \mathrm{H}^1(M, \mathfrak{S}\otimes\mathcal{L}) \longrightarrow \mathrm{L}^2(M, \mathfrak{S}\otimes\mathcal{L})$$

is not invertible (see, e.g., [19, Lemma 2.4 (ii)]). Since the $\mathbb{Cl}_n$-linear spinor bundle $\mathfrak{S}$ is a direct sum of copies of the irreducible spinor bundle $\mathcal{S}$, we obtain that the twisted Dirac operator

$$\mathcal{D}_{\mathcal{L}}\colon\ \mathrm{H}^1(M, \mathcal{S}\otimes\mathcal{L}) \longrightarrow \mathrm{L}^2(M, \mathcal{S}\otimes\mathcal{L})$$

is also not invertible. The functional calculus associated to $\mathcal{D}_{\mathcal{L}}$ restricts as in equation 2.1, hence the operator $T^2 - 1$ is compact for the bounded transform $T \coloneqq \mathcal{D}_{\mathcal{L}}\big(\mathcal{D}_{\mathcal{L}}^2 + 1\big)^{-1/2}$, and we can apply Proposition 3.2. This yields a representation $\rho\colon \pi \to U(d)$ such that $\mathcal{D}_{\mathcal{L}}\otimes_\rho \mathrm{id}$ is not invertible. Note that the representation ρ extends by the universal property of the maximal group C^*-algebra to a representation $\overline{\rho}\colon \mathrm{C}^*\pi \to \mathrm{GL}_d(\mathbb{C})$. We define V as the flat Hermitian bundle $\tilde{M}\times_\rho \mathbb{C}^d$. Following the identifications

$$(\mathcal{S}\otimes\mathcal{L})\otimes_{\overline{\rho}}\mathbb{C}^d \cong \mathcal{S}\otimes\big(\tilde{M}\times_\pi \mathrm{C}^*\pi\otimes_{\overline{\rho}}\mathbb{C}^d\big) \cong \mathcal{S}\otimes V,$$

we obtain $\mathcal{D}_{\mathcal{L}}\otimes_{\overline{\rho}}\mathrm{id} = \mathcal{D}_V$ (compare [18, Proposition 3.1]), hence the twisted Dirac operator $\mathcal{D}_V$ is not invertible. Since the manifold M is closed and the twisting bundle V has finite-dimensional fibers, the spectrum of $\mathcal{D}_V$ consists of eigenvalues only, hence the kernel of $\mathcal{D}_V$ is non-zero. Let u be a non-trivial element in the kernel of $\mathcal{D}_V$. Since the scalar curvature of M is non-negative by assumption, we obtain by the Schrödinger–Lichnerowicz formula

$$\|\nabla u\|_{\mathrm{L}^2}^2 \le \langle\nabla^*\nabla u, u\rangle_{\mathrm{L}^2} + \tfrac{1}{4}\langle\mathrm{scal}_g\, u, u\rangle_{\mathrm{L}^2} = \|\mathcal{D}_V u\|_{\mathrm{L}^2}^2 = 0.$$

This shows that u is parallel, and the second step is proved.

Step 3: There exists a non-trivial parallel spinor on $\tilde{M}$ equipped with the pullback metric. This step corresponds to [18, Proposition 3.3]. The proof is given below for the sake of completeness. Let V be the flat Hermitian bundle over M and u be the twisted parallel spinor constructed in Step 2. We denote by $\tilde{V}$ and $\tilde{u}$ the pullback of V and u to the universal covering of M, respectively. By construction the twisted spinor $\tilde{u}$ is parallel with respect to the pullback connection. Since the covering map $\tilde{M} \to M$ is a local isometry, hence preserves all local structures,

the pullback connection on $\tilde{M}$ equals the usual spinor connection twisted with the pullback connection on $\tilde{V}$. Since V is by definition in Step 2 equal to $\tilde{M} \times_\rho \mathbb{C}^d$ with the representation $\rho\colon \pi \to U(d)$, we obtain

$$\tilde{V} = \tilde{M} \times_{\tilde{\rho}} \mathbb{C}^d \cong \tilde{M} \times \mathbb{C}^d \qquad \text{with} \quad \tilde{\rho}\colon\ \{1\} = \pi_1\big(\tilde{M}\big) \longrightarrow \pi \xrightarrow{\ \rho\ } U(d).$$

This yields an identification $\tilde{\mathcal{S}} \otimes \tilde{V} \cong \big(\tilde{\mathcal{S}}\big)^d$ as bundles with connection, hence the parallel twisted spinor $\tilde{u}$ can be identified with a vector of d parallel spinors on $\tilde{M}$. Since $\tilde{u}$ is non-trivial, at least one component of $\tilde{u}$ is non-trivial and we obtain a non-trivial parallel spinor on $\tilde{M}$. This proves the third step.

Step 4: There exists a finite Riemannian covering of M together with a non-trivial parallel spinor with respect to the pullback spin structure. In the first part of the argument bellow, we follow the approach taken in the proof of [18, Lemma 2.7]. By the structure theorem for Ricci-flat manifolds (see Theorem 3.1), there exists a finite Riemannian covering $(N, g_N) \times (T^q, g_{\text{fl}}) \to (M, g)$ where (N, g_N) is a simply-connected Ricci-flat manifold and T^q the q-torus equipped with a flat metric g_{fl}. This yields that the universal covering of M is isometric to $(N, g_N) \times (\mathbb{R}^q, \tilde{g}_{\text{fl}})$. Note that a Riemannian product admits a non-trivial parallel spinor if and only if each of its factors does. Since the universal covering of M carries a non-trivial parallel spinor by Step 3, (N, g_N) carries a non-trivial parallel spinor. It now suffices to show that a suitable double covering of the q-torus equipped with a flat metric and the spin structure pulled back from M carries a non-trivial parallel spinor.

Since the q-torus is parallelizable and equipped with a flat metric, the $\mathrm{SO}(q)$-principal bundle of orthonormal frames $\mathrm{SO}(T^q)$ can be trivialized such that the trivial connection coincides with the Levi-Civita connection. Let s be the corresponding global section of $\mathrm{SO}(T^q)$. Note that double coverings over a connected manifold X are classified by homotopy classes of homomorphisms $\pi_1(X) \to \mathbb{Z}_2$ [9, Section 1.3].

Let G be $\{0\}$, $\mathbb{Z}$ or $\mathbb{Z}_2$ for $q = 1$, $q = 2$ or $q > 2$, respectively. By the trivialization of $\mathrm{SO}(T^q)$, we obtain an isomorphism $\pi_1(\mathrm{SO}(T^q)) \cong \mathbb{Z}^q \times G$ and the spin structure on T^q, which is a certain double cover of $\mathrm{SO}(T^q)$, is classified by some map

$$(f_1, f_2)\colon\ \mathbb{Z}^q \times G \to \mathbb{Z}_2.$$

For $n > 1$, the condition to be a spin principal bundle yields $f_2 \neq 0$ [13, Section 1, Corollary 1.5]. The isomorphism $\pi_1(T^q) \cong \mathbb{Z}^q$ together with the map $f_1\colon \mathbb{Z}^q \to \mathbb{Z}_2$ gives rise to a double covering $\bar{T}^q \to T^q$. By construction, the pullback of the spin structure of T^q along the map $\bar{T}^q \to T^q$, denoted by $\mathrm{Spin}\big(\bar{T}^q\big)$, is classified by the map $(0, f_2)\colon \mathbb{Z}^q \times G \to \mathbb{Z}_2$. Therefore, the pullback of the section s along the double covering $\bar{T}^q \to T^q$ lifts to a section of $\mathrm{Spin}\big(\bar{T}^q\big)$, hence yields a trivialization of the $\mathrm{Spin}(q)$-principal bundle of $\bar{T}^q$.

Since the trivial connection on the trivialization of $\mathrm{SO}\big(\bar{T}^q\big)$ coincides with the Levi-Civita connection and we just lifted this trivialization, the trivial connection on the trivialization of $\mathrm{Spin}\big(\bar{T}^q\big)$ also coincides with the Levi-Civita connection. It follows that the constant sections of the associated trivialized spinor bundle over $\bar{T}^q$ are parallel, and Theorem A is proved. □

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